UTN

University of Technology Nuremberg

Rating Plausibility of Word Senses in Ambiguous Sentences through Narrative Understanding

SemEval 2026 Task 05

Costache Rebeca, Cezara Iosif, Alex Neagu

Word Sense Disambiguation (WSD)

- Word Sense Disambiguation is a graded, non-binary problem
- Real word meanings are 'more likely' or 'less likely' based on context.
- AmbiStory task requires deep narrative understanding from surrounding text (5 sentences of context).
- The goal is predicting human plausibility on a scale from one to five.

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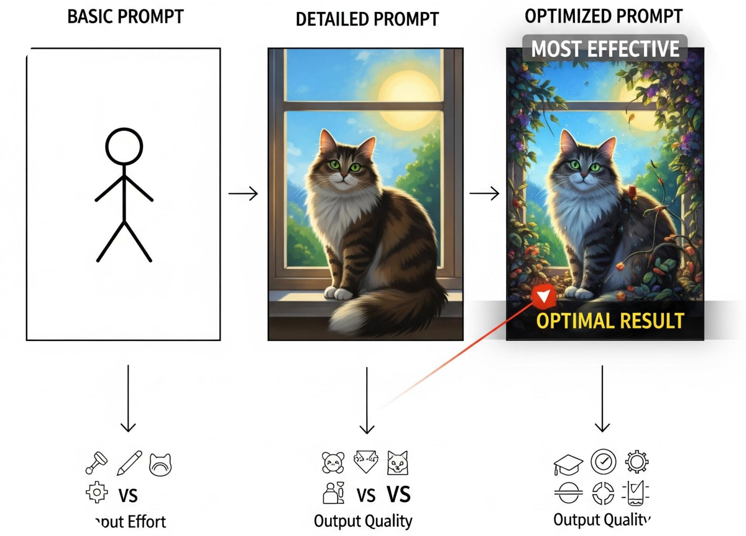
"homonym": "bugs",
"judged_meaning": "general term for any insect or similar creeping or crawling invertebrate",
"precontext": "Anna was having a tough week. Her room was a mess, and her computer kept crashing.
Frustrated by everything going wrong, she called Jen.",
"sentence": "She asked her friend to help her get rid of the bugs.",
"ending": "They were crawling on the keyboard. Maybe that was the reason it didn't work."
```

Why Encoders Work Best

- Encoder-only models are better suited for judging plausibility.
- The task is discriminative, requiring evaluation rather than generating new text.
- Encoder models analyze the entire story context bi-directionally at once.
- Generative models, focused on next-word prediction, can miss overall narrative flow.

Prompting Strategies



- The Rich Input Just the text and the meaning separated by tags. (Result: The model didn't know what to focus on).

- Instructional Prompt Asking the model a direct question. (Result: Better, because the model understood the "job").

- Masked Prompt Replacing the word with [MASK]. Explain that this was your "secret weapon"—it forced the model to look at the story context because the word itself was gone.

Results and Analysis

01 —— The Simple Prompt was the clear winning strategy (spearman validation: 0.53)

02 —— The Masked Prompt was second best (spearman validation: 0.51)

03 —— The 4-bit Mistral ensemble performed poorly (spearman validation: 0.0332) due to issues regarding memory and time resources

04 —— CUDA Out-of-Memory errors limited training on the Kaggle T4 GPUs.

Final Competition Results

- Final competition submission used DeBERTa-base with Masked Input.
- This reliable version achieved a 0.65 Spearman Correlation score on the test dataset.
- Simple prompt came close, with a score of 0.64
- Smart prompt strategies outperformed large, expensive ensemble models.